

Nima Saghi

(+46)736353818

nima.saghi1993@gmail.com

[LinkedIn](#) Profile

Github: <https://github.com/NimaaaAI>

Artificial Intelligence and Automation Engineer with a strong background in applied machine learning and deep learning, spanning computer vision, speech systems, and LLM-based applications. Experienced in translating research into practical AI solutions, from anomaly detection and multi-task learning to deployed chatbot and decision-support systems.

PROFESSIONAL EXPERIENCE

Integration Developer

(Artificial Intelligence – Machine Learning & Deep Learning)

Trollhättan Stad

Trollhättan, Västergötland

February 2025–December 2025

- **Speech Transcription System:** Built a fully local speech-to-text pipeline using open-source models.
- **Text-to-Speech (TTS) Engine:** Developed a local TTS service generating natural-sounding voice output.
- **AI Translation Model:** Implemented an on-premises multilingual neural machine translation system.
- **Local Chatbot & Orchestrator:** Created a local chatbot with intent detection, planning, MCP-integrated APIs, and fine-tuned embeddings for database understanding.
- **Employee Absence Forecasting:** Built an ML model connected to internal databases with scheduled retraining and API access through the chatbot.
- **Large-Scale RAG System:** Developed a nightly-ingested RAG pipeline with embeddings, vector indexing, and chatbot integration.
- **Interactive Voice AI Assistant:** Built a phone-based AI agent using local STT/TTS and LLMs for real-time voice conversations.
- **QGIS & Deep Learning Object Detection:** Used QGIS and deep learning models to analyze geospatial images and perform object detection.

Thesis Work

(Road Anomaly Detection and Severity Estimation)

Volvo Cars

Göteborg, Västergötland

January 2025–June 2025

- Develop and implement ML and DL models for anomaly detection.
- Apply Image Processing techniques to enhance detection accuracy.
- Conduct experiments, model evaluation, and performance optimization.

Research Assistant

(Artificial Intelligence Pipelines)

Högskolan Väst

Trollhättan, Västergötland

June 2024–August 2024

- **Multi-Task CNN for Additive Manufacturing:** Built a custom CNN with manual layers for defect classification and laser-path regression, combining both tasks into a single multi-task learning model.
- **Reinforcement Learning for Heat-Camera Automation:** Developed an RL-based model for a thermal camera system to identify when welded layers in additive manufacturing were ready for the next step based on heat patterns.
- **Remote GPU Computing Environment:** Implemented a multi-GPU virtual environment with centralized dependencies and VS Code access, enabling students to run heavy workloads remotely without setup issues.

Conducted electronic calibration of laboratory equipment and meticulously reported results in accordance with relevant standards and instructions.

Provided technical and quality assessment services, ensuring laboratory equipment met rigorous performance and safety standards.

EDUCATION

University West	Trollhättan, Västergötland
Master in Artificial Intelligence and Automation	2023–2025
Azad University Yadegar-e-Imam Khomeini	
Bachelor of Science in Electronics Engineering Technology	2013–2015
Associate’s degree in Electronics Engineering Technology	2010–2013

AI Published Paper: Marine Debris Detection from Sentinel-2 DOI:[10.1109/RADIOELEKTRONIKA65656.2025.11008372](https://doi.org/10.1109/RADIOELEKTRONIKA65656.2025.11008372)

ADDITIONAL SKILLS

Programming Languages:

- Python, C++, MATLAB, RAPID, PLC, Bash/Shell Scripting, SQL.

AI, ML & Data Science Tools:

- Deep Learning & Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, OpenAI Gym, Vision Transformers, CNNs, RNNs, Transformers, Ollama, HuggingFace (LLMs, embeddings, fine-tuning), ONNX, NumPy, SciPy, Pandas, scikit-image, Pillow, Medica images.
- LLMs & NLP: LangChain, RAG pipelines, vector databases (Qdrant, Chroma), prompt engineering, model fine-tuning, quantization, optimization, deployment (local + containerized).
- Computer Vision: VLMs, OpenCV, image processing, deep learning–based detection/classification, thermal imaging, QGIS for geospatial AI tasks.
- Visualization: Matplotlib, Seaborn, Plotly.

Development, Deployment & System Tools:

- MLOps & Environment Management: Google Cloud Platform, Vertex AI Studio, Conda, venv, Docker (containerization), Kubernetes (orchestration), CI/CD pipelines.
- Backend & API Development: FastAPI, Flask, REST APIs, MCP-based orchestration.
- Hardware & Automation Tools: TwinCAT, Proteus, Altium Designer, industrial camera systems, GPU optimization and setup for training/inference.
- Databases & Data Handling: MongoDB, MySQL, vector DBs, data pipelines.
- Version Control & Collaboration: Git, GitHub, GitLab.
- Interactive Development Environments: Jupyter Notebook, JupyterLab, Anaconda, VS Code.

SOFT SKILLS & Languages

- Analytical Thinking - Project planning - Collaboration in teams - Adaptability to new tools - Clear technical communication
- English (Fluent), Swedish (Basic)
- Swedish Driving’s License: B

REFERENCES

References available upon request